

MT 508: Iron and Steel Making

Course Code	MT 508
Course Name	Iron and Steel Making
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	B.Tech./PG/Ph.D.
Mutual Exclusion	None
Approval	57th BoA (Oct 2024)

Course Contents

- **Introduction to Iron Making:** Introduction: Various routes of modern steel making (BF-BOF, DRI-EAF, SR). Blast furnace Ironmaking: Shape of the furnace, Various reactions, and zones in the blast Furnace Thermodynamics of Iron making, Direct & indirect reduction Blast furnace as an efficient counter-current gas-solid reactor with maximum carbon efficiency. [6 Hours]
- **Heat and Mass Transfer in the blast furnace:** Overall heat and material balance in the blast furnace. Rist diagram based on oxygen balance. Rist diagram based on heat & material balance. Blast furnace as a two-stage reactor. A predictive model for coke rate in the blast furnace. [8 Hours]
- **Transport processes in a blast furnace:** Aerodynamics in Blast Furnace (pressure drop, fluidization, channeling, flooding). Burden preparation (sintering, pelletization, coke making). Testing of raw materials. Burden distribution in blast furnace. Auxiliary injection, fuel efficiency, and productivity in blast furnaces. [6 Hours]
- **Steelmaking Fundamentals:** Thermodynamics of steelmaking. Introduction to LD steel making (process, emulsion, decarburization, catch carbon technique, impurity & slag evolution). Hybrid steel making. Modern EAF/IF steel making. Secondary Steelmaking: Thermodynamics & kinetics of deoxidation. Secondary steel making: Vacuum treatment of steel and processes, gas stirring. [8 hours]
- **Secondary Steelmaking and Alternative Ironmaking Routes:** Secondary Steelmaking: Ladle desulphurization, inclusion modification by calcium treatment. Inclusion control by optimizing upstream operating parameters. Heat transfer and segregation during casting. Ingot casting, continuous casting, defects in steel. Environmental issues related to ironmaking; Alternative routes of iron & steel making (rotary kiln, Corex, Midrex). [10 Hours]

Laboratory Modules

1. NA

Text Books

1. H.S. Ray and A. Ghosh, **Principles of extractive metallurgy**, Wiley Eastern Ltd., 1991.
2. Dr. R. H. Tupkary, **Modern ironmaking**.
3. R.W. Backforth, **Manufacture of Iron & Steel**, Vol. I.
4. G. Word, **Physical Chemistry of Iron & Steel**.
5. A.K. Biswas, **Principals of Blast Furnace Ironmaking**.
6. Dr. R.H. Tupkary, **Modern Steelmaking**.
7. E.T. Tukdogan, **Fundamentals of steel making**.

References

1. Turkdogan, E.T., and Fruehan, R.J., **Fundamentals of iron and steelmaking**.
2. RJ Fruehan, ed., **The Making, Shaping and Treating of Steel**, Steelmaking and Refining Volume 11, 11th Edition, AISE Steel Foundation, pp. 125-126, 1998.
3. A. Ghosh, **Secondary processing and casting of liquid steels**.